

PRODUCT ANALYTICS CASE STUDY

NordicFlow CRM

From Business Problem to Actionable Insight — A Raw-to-Dashboard Analytics Build

PYTHON

SQL

POWER BI

A end-to-end analytics build: Raw data → Python cleaning → SQL gold layer → Power BI dashboard

Agenda

01

The Business Problem

Who NordicFlow is, and the challenges pushing them toward product analytics

02

Our Approach

How raw data became a governed, decision-ready dashboard — Python, SQL, Power BI

03

Findings

What the data actually shows about accounts, activation, channels and churn

04

Recommendations & Conclusion

What NordicFlow should do next, grounded in the evidence

PROBLEM STATEMENT

Who is NordicFlow CRM?

NordicFlow CRM is a Copenhagen-based, European B2B SaaS company that sells a subscription CRM platform to small and mid-sized businesses (roughly 20–300 employees).

Its market position: a GDPR-first alternative to US platforms — competing on usability, workflow design, and compliance rather than feature parity with global incumbents.

The core tension: logging in and clicking around isn't the same as running your sales process through NordicFlow. Activity is not value — and that distinction shapes every metric in this project.

AT A GLANCE

HQ

Copenhagen, Denmark

FOUNDED

2019

PRODUCT

Subscription B2B CRM

CUSTOMER SIZE

~20–300 employees

POSITIONING

GDPR-first, privacy-led

KEY INDUSTRIES

SaaS, Prof. Services, Logistics,
Manufacturing, B2B Agencies

Four Challenges Driving This Project

1

Inconsistent Activation

Trial signups are climbing steadily, but a meaningful share of new accounts never reach real day-to-day usage.

2

Activity Mistaken for Value

Users log in without touching pipeline, deal progression, or automation — surface activity looks like engagement, but isn't.

3

Early Churn

Churn is emerging among smaller customers, usually because they fail to realize value quickly enough — not because of missing features.

4

Conflicting Dashboards

Different teams pull different numbers from the same data, making it hard for leadership to prioritize or measure impact.

PROBLEM STATEMENT

The Objective

Turn fragmented raw data into a single, trusted view of how customers actually adopt NordicFlow — so leadership can act on evidence instead of opinion.

- Define trusted, consistent metrics for activation, engagement and churn
- Make assumptions explicit so results are interpreted the same way across teams
- Explain why activation and churn happen — not just report the numbers
- Deliver a governed data pipeline that any team can query with confidence

SCOPE OF THIS BUILD

116

Accounts analyzed

352

Active users tracked

4

Fact tables (Gold layer)

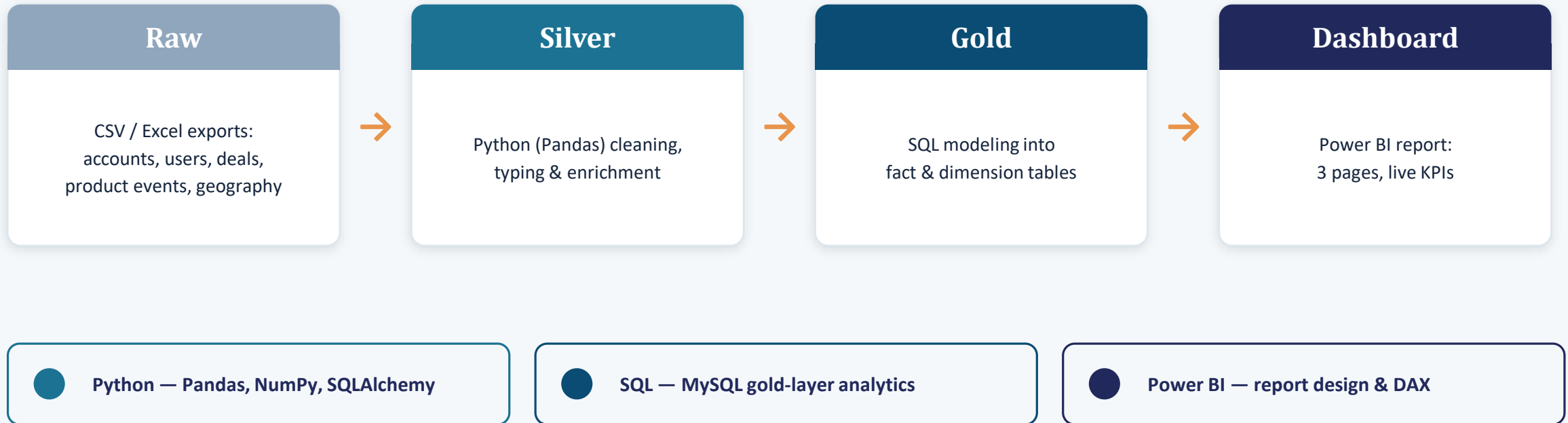
3

Power BI report pages

OUR APPROACH

One Pipeline, Three Tools

Raw exports were carried through a Bronze → Silver → Gold model before ever reaching a chart — each stage adding trust, not just transformation.



Cleaning & Shaping the Data in Python

Four raw sources — accounts, users, deals, and product events — were profiled and cleaned with Pandas before loading into the Silver layer (MySQL).

1

Profile & Inspect

- Checked shape, dtypes, nulls
- Reviewed distributions with `.describe()`
- Dropped records missing key fields (e.g. industry)

2

Standardize

- Trimmed whitespace in text fields
- Unified category labels (product-led → product_led)
- Normalized inconsistent country codes

3

Enrich & Model

- Parsed date fields (trial, deal, activity dates)
- Derived trial length, sales-cycle days, recency
- Joined accounts to a geography lookup table

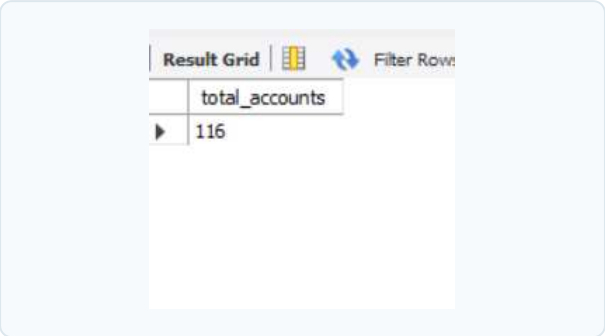
4

Load to Silver

- Pushed clean frames to MySQL via SQLAlchemy
- Built `silver_dim_accounts / _users / _deals`
- Built `silver_product_event` for behavioral data

SQL: Establishing the Core KPIs

Once data landed in the Gold layer (gold_fact_accounts, gold_fact_users, gold_fact_product_event), SQL queries produced the platform's headline numbers.

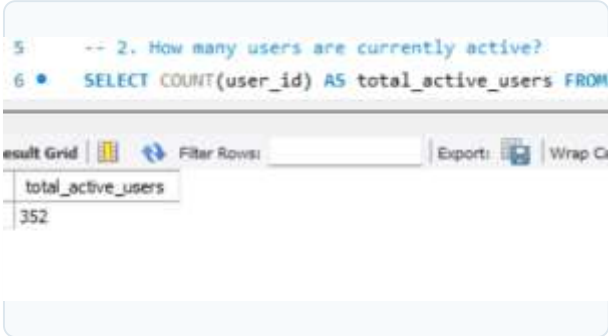


total_accounts
116

QUERY 1 · TOTAL ACCOUNTS

116

Registered accounts on the platform



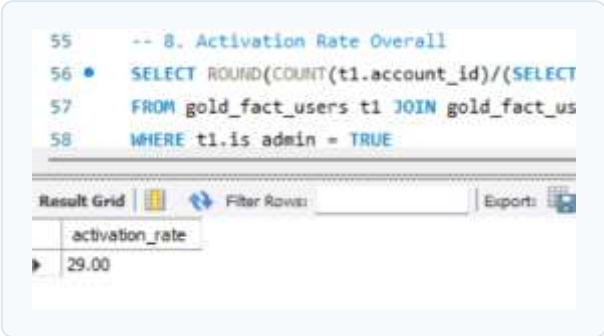
```
-- 2. How many users are currently active?
SELECT COUNT(user_id) AS total_active_users FROM
```

total_active_users
352

QUERY 2 · ACTIVE USERS

352

Users currently active



```
-- 8. Activation Rate Overall
SELECT ROUND(COUNT(t1.account_id)/(SELECT
FROM gold_fact_users t1 JOIN gold_fact_us
WHERE t1.is_admin = TRUE
```

activation_rate
29.00

QUERY 8 · ACTIVATION RATE

29%

Overall admin-activation rate

SQL: Segmenting Accounts

Grouping and window-style aggregation queries broke the account base down by industry and geography to spot where volume — and risk — concentrates.

```
-- 3. Number of accounts based on each industry
SELECT industry, COUNT(account_id) AS account_count
FROM accounts
GROUP BY industry
```

industry	account_count
Technology Services	24
SaaS	22
Manufacturing	20
Wholesale	18
B2B Agency	18
Logistics	17
Professional Services	10

Query 3 — Accounts by Industry

```
-- 4. Top 5 Countries by Account Volume
SELECT COALESCE(country_name, country_code) AS country,
COUNT(account_id) AS total_accounts
FROM accounts
GROUP BY country
ORDER BY total_accounts DESC
LIMIT 5
```

country	total_accounts
France	26
Sweden	23
Denmark	21
NL	16
United Kingdom	15

Query 4 — Top 5 Countries by Volume

```
-- 5. Active Accounts & Churn Rate by Country
SELECT COALESCE(country_name, country_code) AS country,
COUNT(account_id) AS total_accounts,
SUM(CASE WHEN status = 'Active' THEN 1 ELSE 0 END) AS active_accounts,
SUM(CASE WHEN status = 'Churned' THEN 1 ELSE 0 END) AS churned_accounts,
SUM(CASE WHEN status = 'Churned' THEN 1 ELSE 0 END) / COUNT(account_id) AS churn_rate
FROM accounts
GROUP BY country
```

country	total_accounts	active_accounts	churned_accounts	churn_rate
Sweden	23	19	1	4.35
Denmark	21	18	2	9.52
NL	16	14	0	0.00
United Kingdom	15	14	1	6.67
Germany	14	10	1	7.14
Norway	14	13	1	7.14
France	13	10	2	15.38

Query 5 — Active vs. Churned by Country

Technology Services and SaaS lead by industry volume, while France, Sweden and Denmark are the three largest markets — and two of them also carry the highest churn (see next slides).

SQL: What Drives Activation?

Joining accounts to product-event activity isolated which acquisition channels and industries actually convert signups into activated, admin-configured accounts.

```
33 -- 6. Which customer acquisition channels yield the highest activation accounts?
34 • SELECT
35     a.acquisition_channel,
```

acquisition_channel	activated_accounts
content	25
paid_search	20
partner	23
product_led	20
sales_led	28

Query 6 — Activated Accounts by Acquisition Channel

Sales-led and content channels activate the most accounts. Activation rate varies sharply by industry — SaaS converts best, Professional Services and B2B Agency lag furthest behind.

```
43
44 -- 7. Activation Rate by industry
45 • WITH cte AS(
46     SELECT t2.industry, COUNT(t1.a
```

industry	activation_rate
Wholesale	1.500
Technology Services	1.000
SaaS	2.000
Manufacturing	1.300
B2B Agency	0.500
Logistics	0.500
Professional Services	0.300

Query 7 — Activation Rate by Industry

Bringing It Together in Power BI

The gold-layer tables feed a 3-page Power BI report — each page built for a different audience and question.

PAGE 1

Executive Overview

Leadership snapshot

VISUALS ON THIS PAGE

- KPI cards
- Trend (area)
- Column chart
- Donut / Pie

PAGE 2

Product Usage & Adoption

Which features actually get used

VISUALS ON THIS PAGE

- Bar chart
- Pivot table
- Trend (area)
- Treemap

PAGE 3

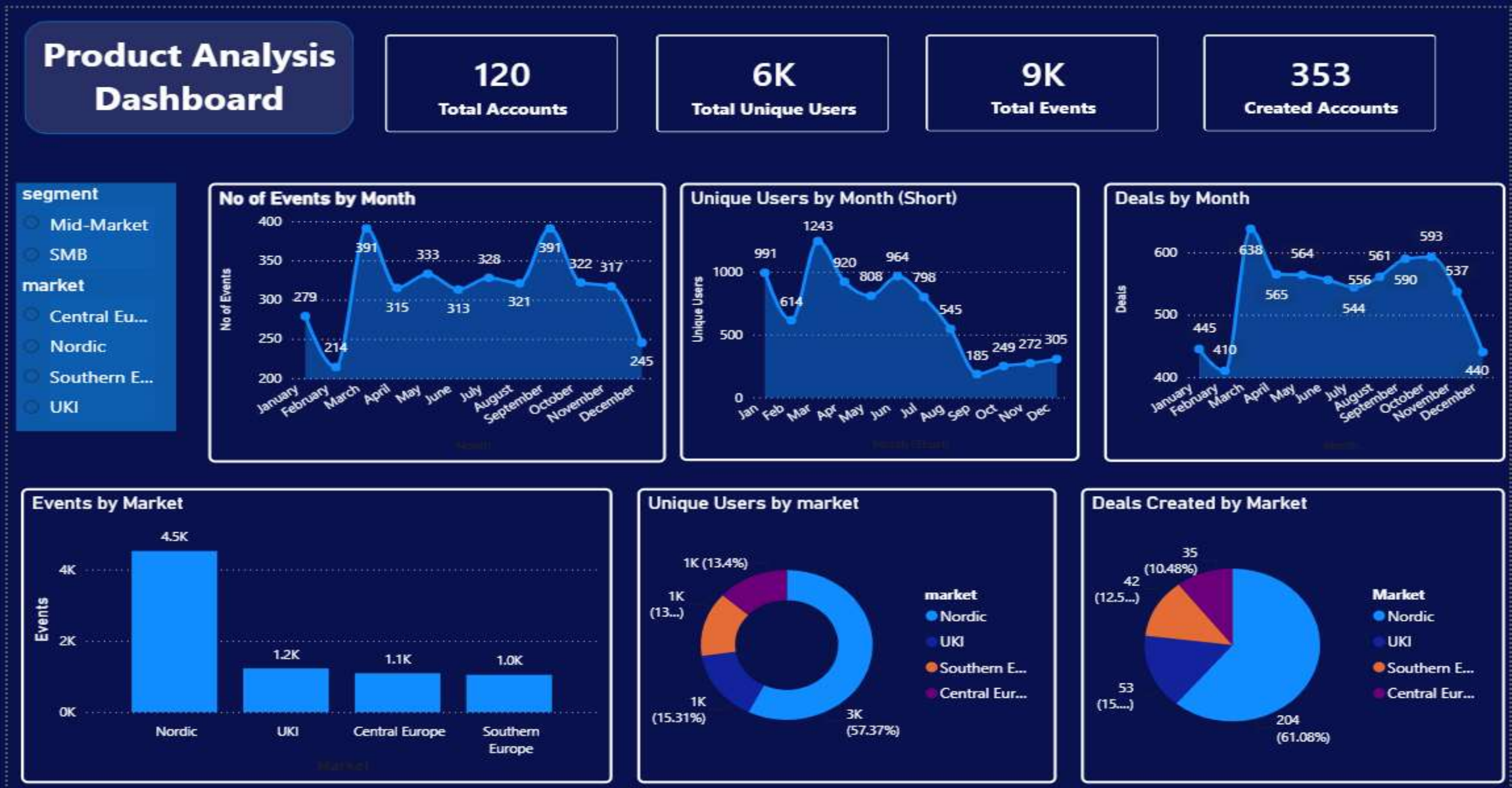
Activation & Time-to-Value

Speed and depth of onboarding

VISUALS ON THIS PAGE

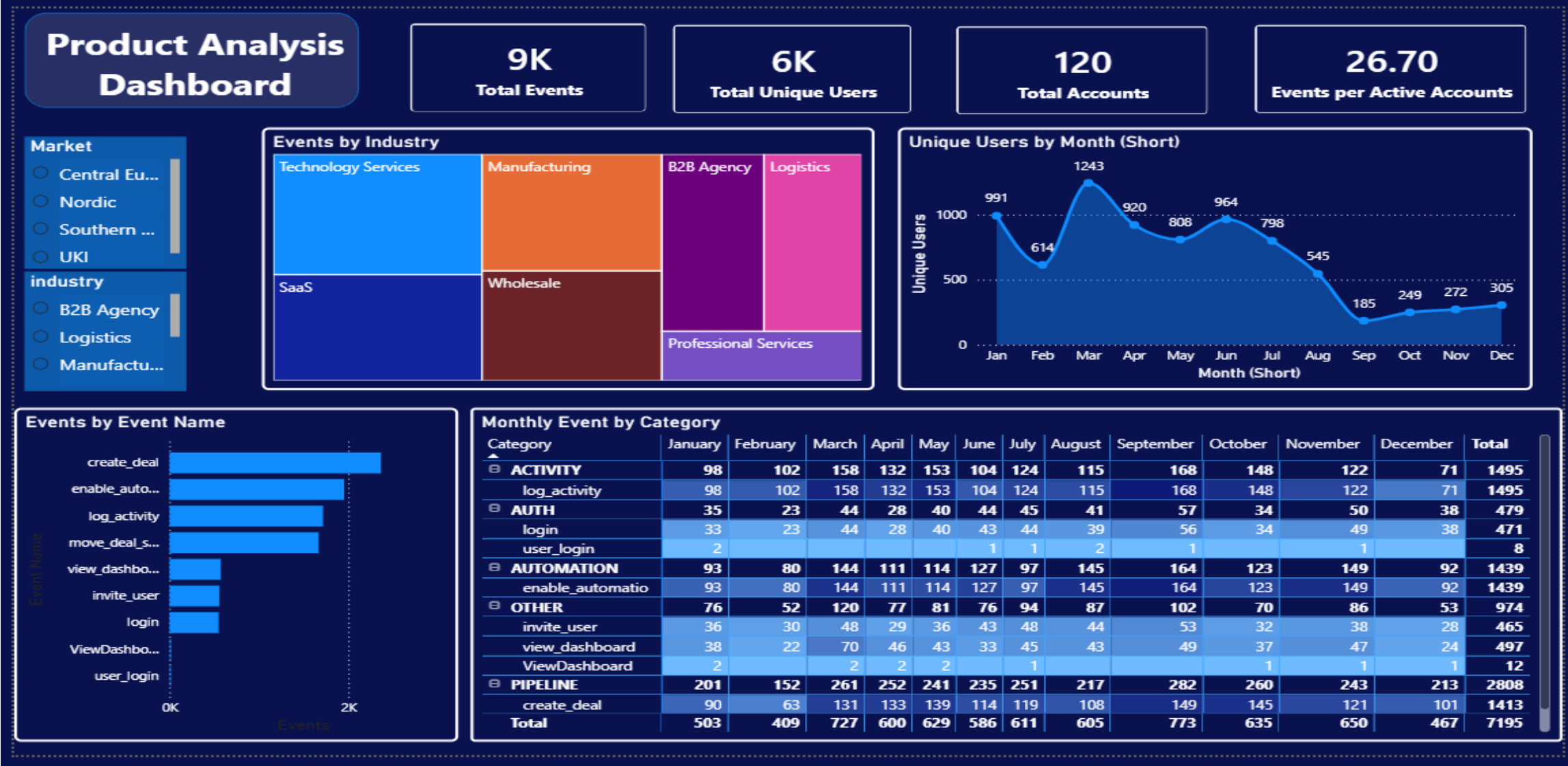
- KPI cards
- Trend (area)
- Column + Pie
- Treemap

Executive Dashboard Snapshot



FINDINGS

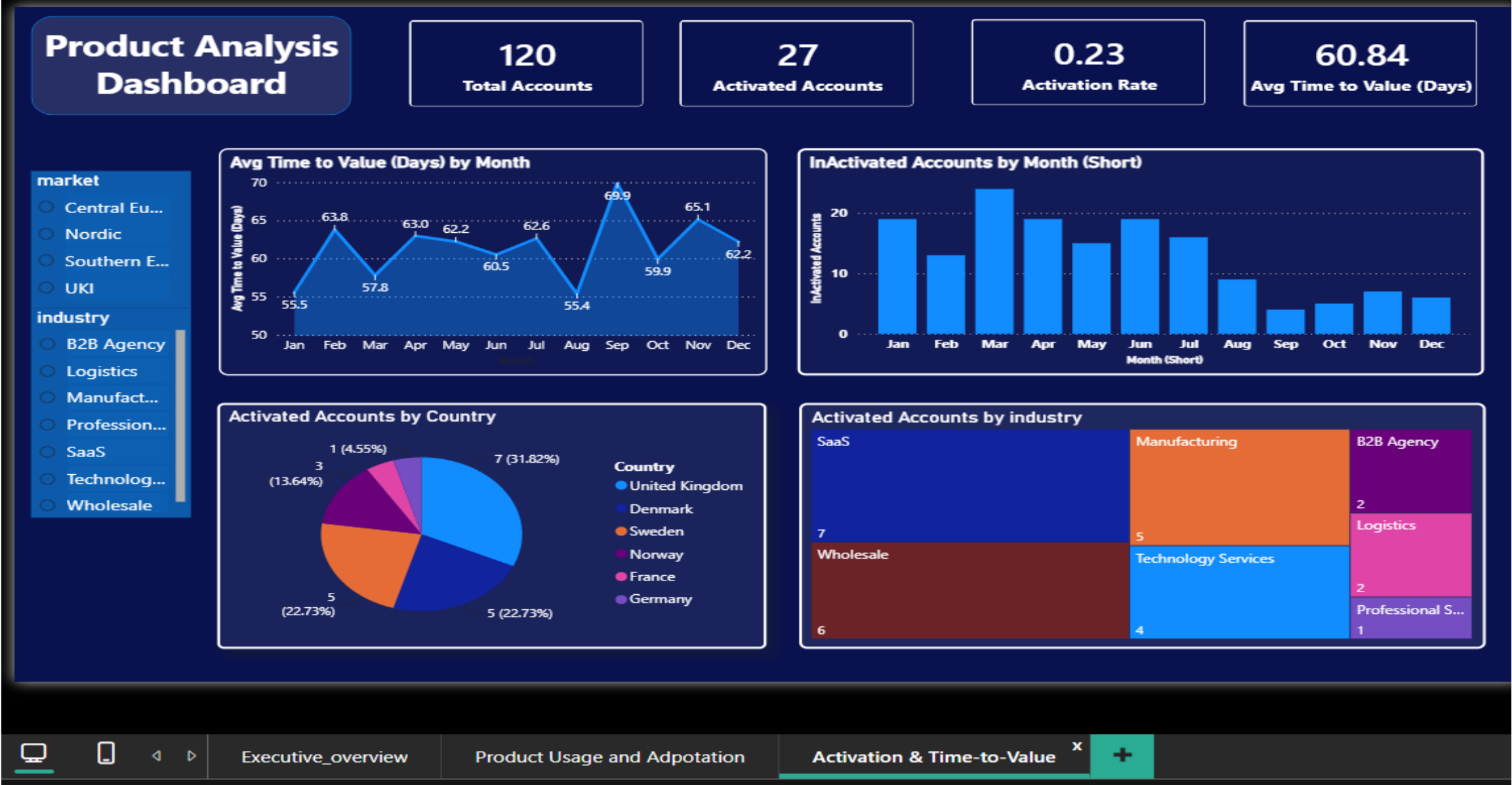
Where Activation and Churn Diverge



France and Denmark combine high account volume with the highest churn rates (15.4% and 9.5%) — while sales-led and content channels activate meaningfully more accounts than paid search or product-led signup.

NordicFlow CRM | Product Analytics

FINDINGS



RECOMMENDATIONS

What NordicFlow Should Do Next

Redesign onboarding for low-activation industries

Professional Services (0.3%) and B2B Agency (0.5%) lag far behind SaaS (2.0%). Build industry-specific onboarding playbooks around their most common workflows.

Double down on sales-led and content acquisition

These channels activate the most accounts (28 and 25). Shift growth budget and rep time toward them before scaling paid search or self-serve product-led signup.

Launch a proactive save motion in France and Denmark

The two highest-churn markets (15.4%, 9.5%) also carry large account volumes. Trigger Customer Success outreach the moment early usage signals stall.

Redefine “engagement” around core actions

Track pipeline configuration, deal progression and automation — not logins — as the activation signal, so every team reports the same number.

Establish one governed source of truth

Standardize entities like country codes and channel labels at ingestion (this project found join mismatches from inconsistent codes) so every dashboard agrees.

CONCLUSION

From Raw Data to a Decision-Ready Story

This project turned scattered CSV/Excel exports into a governed Python → SQL → Power BI pipeline, and the resulting numbers tell a clear story: NordicFlow's activation problem is uneven across industries and channels, and its churn risk is concentrated in a few key markets — both fixable with targeted, not company-wide, action.

NEXT STEPS

- Extend the fact tables with historical cohorts to trend activation and churn over time
- Automate the Silver → Gold refresh so the dashboard stays current without manual reruns
- Roll the Executive Overview page out as the single shared source of truth across teams

PROJECT AT A GLANCE

116

Accounts

352

Active Users

29%

Activation

7.2%

Avg. Churn